

# An Overview Of Recommendation Systems

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# Overview

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# Content-Based vs. Collaborative Filtering

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  - Focused on relationships between users and items
  - Recommendations made on similarity of ratings of other shared items

# Pros and Cons

- Content-Based:
  - Pro: Simplicity - just need descriptions and specs of the items and information about users.
  - Pro: Data is generally easier to curate.
  - Con: Difficult to make recommendations on rarely purchased items.

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  - Con: Difficult to make recommendations on rarely purchased items.
- Collaborative Filtering:
  - Pro: Provides personalized recommendations to the user.
  - Pro: Possible to make good recommendations in sparse situations.
  - Con: Takes time to collect a critical mass of user reviews.
  - Con: More complex and compute-intensive.



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The point is to estimate the null values of  $U$  to accurately predict what a user is likely to buy and enjoy.

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- Demographics
- Features of items prevalent in their purchases

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  - Can use LSH methods from last time to narrow down items likely to have high similarity.
- Classification methods:
  - Decision trees/random forest/gradient boosted trees
  - Many of these methods are tractable even for large user-bases/item sets.

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Clustering may be applied to users and items to reduce the size of the utility matrix.



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Non-negative in this context means  $AB$  must have nonnegative entries.